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ICE 2.2



Aman Gupta • Feb 5, 2025 (Edited Feb 15, 2025)

8 points

1. (2 marks) Using Caesar cipher on English text, with key = 4, what is the cipher corresponding to IAMOK?

- KOMAI
- **MEQSO (correct)**
- LDPRN
- None of the above

2. (2 marks) What is the computational effort to launch a brute force attack on a cipher that uses permutation-based substitution of English text with 26 characters?

- **$O(26!)$ (correct)**
- $O(2^{26})$
- $O((26^2)^2)$
- None of the above

3. (2 marks) We will use a substitution cipher, where key K is a permutation of the symbols. The symbols consist of {0, 1, 2}. The key is 0-> 1, 1->0, 2->2. What is the resulting ciphertext corresponding to plaintext 01221.

- 01221
- 22010
- **10220 (correct)**
- None of the above

4. (2 marks) We will try to launch a brute-force attack to determine the link for a Google Meet that is encrypted using a substitution cipher, where key K is a permutation of the symbols in {a, b, ..., z}? A Google Meet link is for example, meet.google.com/trc-xwqt-iks. The string 'meet.google.com' and the characters '.', '/', '-' are not encrypted. Will we ever succeed? Or will we always fail?

- Yes, we will succeed always.
- Yes, we will succeed sometimes, but not always.
- **We will not succeed, given that there is no way to determine if we have got it right. (correct)**
- We will never succeed since we are not encrypting the string 'meet.google.com' and the characters '.', '/', '-'.
- None of the above



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